

ME5412 Robotics for Healthcare

Module briefing

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Module Syllabus and Schedule

Week	Date	CA	Topic	Lecturer
Week 1	11/8/2026		Course briefing, introduction to healthcare robotics, principles of Assistive Technology	Cecilia Laschi
Week 2	18/8/2026		Interfaces for assistive devices, mobility aids	Cecilia Laschi
Week 3	25/8/2026		Manipulation aids, upper limb prostheses	Cecilia Laschi
Week 4	1/9/2026		Introduction to stroke and stroke rehabilitation	Haoyong Yu
Week 5	7/9/2026		Lower limb rehabilitation robotics	Haoyong Yu
Week 6	15/9/2026		Upper limb rehabilitation robotics	Haoyong Yu
Recess				
Week 7	29/9/2026		Project assignment	Both
Week 8	6/10/2026	50%	Mid-Term Quiz (1h30)	Both
Week 9	13/10/2026		Project time and consultation	Both (on demand)
Week 10	21/10/2026		Project time and consultation	Both (on demand)
Week 11	27/10/2026		Project time and consultation	Both (on demand)
Week 12	3/11/2026	50%	Project presentations	Both
Week 13	10/11/2026		Project presentations cntd.	Both

Quiz (assessed, 50%)

- Online quiz on **ExamSoft***
- In person, open book, 1h30 duration, October 6th, 2026
- Questions on the topics of the lectures, open book

*Students' **Briefing Sessions:** <https://nus.atlassian.net/wiki/x/O4BXAQ>

Self-learning: briefing slides or [Exemplify Student Briefing Video](#)

Student guides:

- Digital Assessment Student
Wiki: <https://nus.atlassian.net/wiki/spaces/DAsstudent/overview>
- Exemplify Student Briefing Slides: <https://nus.atlassian.net/wiki/x/M4BXAQ>
(Refer to In-Person Assessment)
- Download and Install Exemplify: <https://nus.atlassian.net/wiki/x/HYBXAQ>
- How do I login to Exemplify? <https://nus.atlassian.net/wiki/x/7YBXAQ>
- Common Briefing Sessions: <https://nus.atlassian.net/wiki/x/O4BXAQ>
- Practice Exams: <https://nus.atlassian.net/wiki/x/IIBXAQ>

Project Assignment (assessed, 50%)

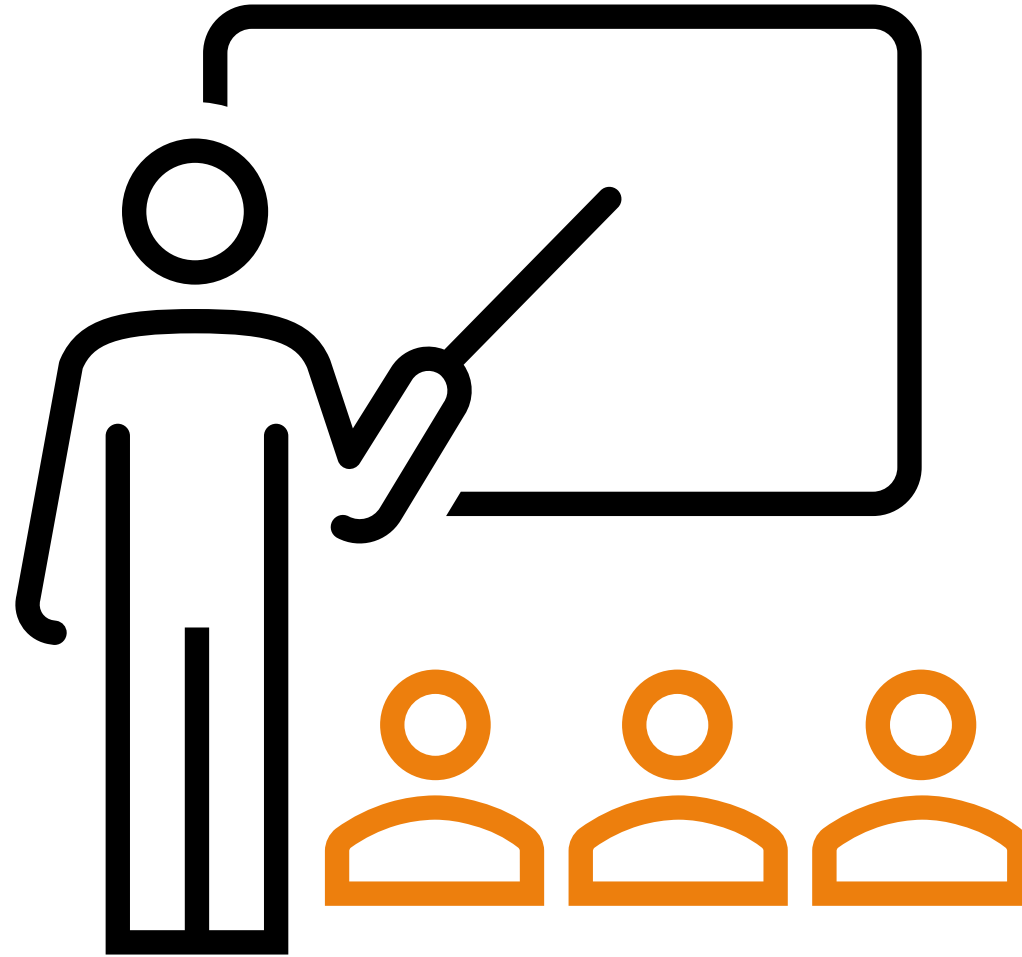
- Groups* of 5 students
- Each group free to choose a topic, within the course scope
- Prototypes not expected, but concept/design/simulation
- Presentation by each group, to the other students**, in 10-15 min
- Submission of slides as a report, by each group, November 3rd, 2026

*Please organize yourselves in groups and start setting up the groups in Canvas → ME5412 → People → Groups → ME5412 Project.

** All students are supposed to attend all presentations.

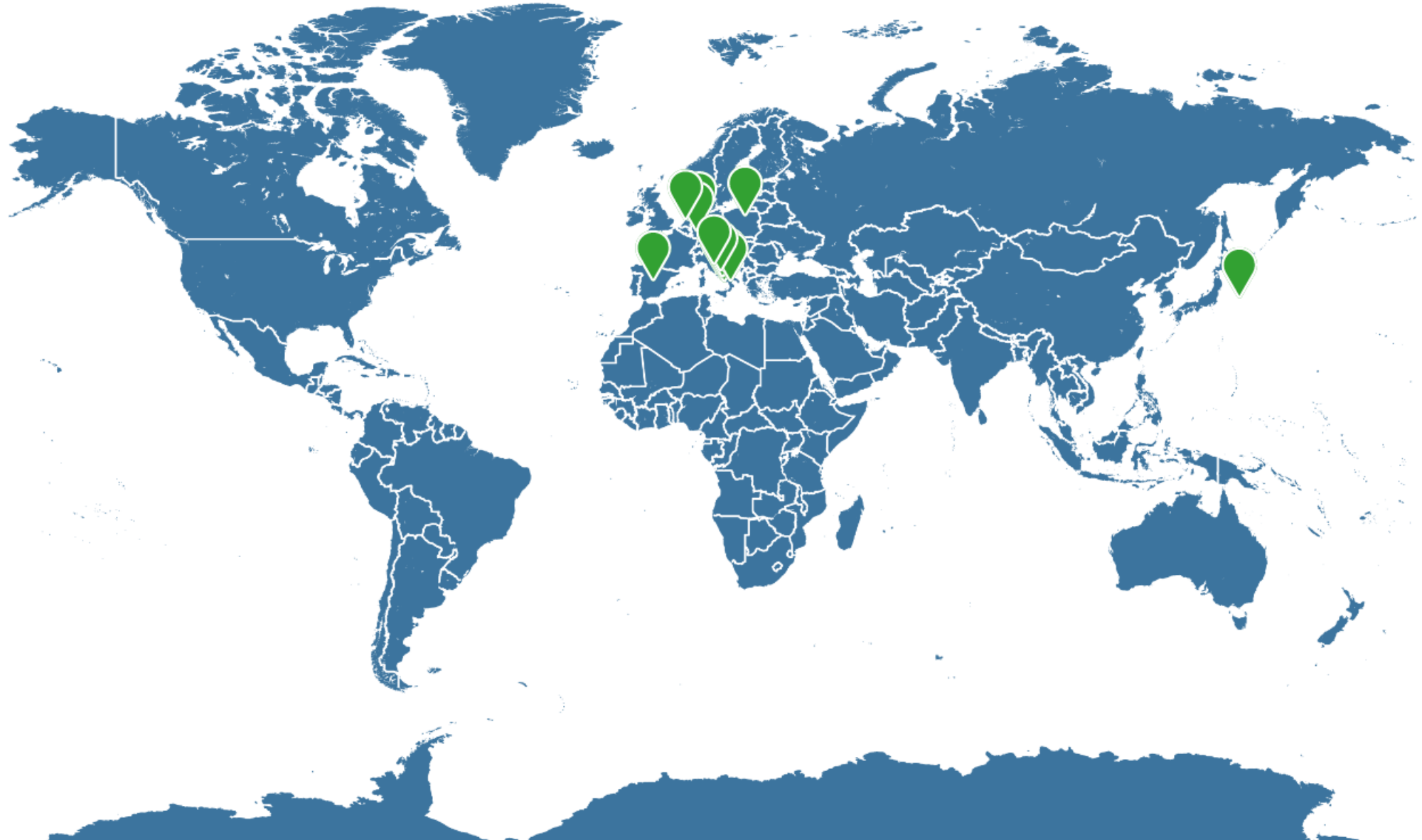
Pollev.com/ceciliaschi

Your inputs



Where are you from?

15



ME5412 Robotics for Healthcare

Introduction to Healthcare Robotics

Cecilia Laschi

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Robots for Healthcare

- Why healthcare robotics, why now
- What counts as a "healthcare robot" — and the autonomy spectrum
- A taxonomy: five families of healthcare robots
- Core technical challenges across all five
- Regulatory and ethical landscape
- Course roadmap

Why Healthcare Robotics, Why Now

- Three converging pressures are pulling robotics out of the lab and into the clinic



2.1B

people worldwide will be aged 60+ by 2050 (WHO)



10M

projected global health-worker shortfall by 2030 (WHO)



~35%

of a nurse's shift spent on non-clinical tasks that robots can offload

- Robotics won't replace clinicians — it extends scarce clinical time, reduces physical strain, and enables care that scales with demand.

A Growing Healthcare Robotics Market (2025-2035)

FORECAST PERIOD

2025-2035

CAGR

14.6%

2025 MARKET SIZE

USD 16.55 Billion (2025)

2035 MARKET SIZE

USD 65.29 Billion (2035)

Two converging forces:

1. accelerating integration of machine intelligence into surgical workflows
 2. wave of reimbursement reforms that are making capital-intensive robotic platforms financially accessible to mid-tier healthcare facilities.
 - The U.S. Centers for Medicare & Medicaid Services (CMS) expanded coverage codes for robot-assisted procedures in 2024,
 - The European Commission allocated EUR 1.2 billion under Horizon Europe for next-generation clinical automation.
- North America Medical Robotics Market expected to hold 38.1% revenue share, due to well-established reimbursement routes and a high concentration of teaching hospitals.
 - Asia-Pacific is the fastest-expanding market with a CAGR of 16.1%. Government-supported rehabilitation initiatives in China, India and Japan drive growth.
 - Europe ranks second with a 27.5% share, bolstered by unified MDR regulatory clearances and cross-border surgical training consortia.
 - The ability of manufacturers to implement predictive analytics into platform ecosystems will define the next decade.

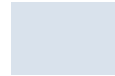
What Counts as a "Healthcare Robot"?

WORKING DEFINITION

A sensing–reasoning–acting system that autonomously or semi-autonomously performs physical tasks within clinical, rehabilitative, or care-related settings, in direct or indirect interaction with patients, clinicians, or care environments.

- **Sense** Perceives patients, tools, or environment
- **Reason** Plans actions under uncertainty & safety limits
- **Act** Physically manipulates, moves, or assists

SPECTRUM OF AUTONOMY



Tele-operated

Human drives every motion (e.g. remote surgery console)

Supervised / shared control

Robot assists, filters tremor, enforces safety zones

Task autonomy

Robot executes a defined task, human sets goals

Full autonomy

Robot perceives, decides, and acts with no human in the loop

A Taxonomy of Healthcare Robots



Do you know robots in the 5 categories?

0

Surgical & Interventional

Rehabilitation & Assistive

Hospital Service & Logistics

Socially Assistive & Companion

Diagnostics & Monitoring

Surgical & Interventional Robots

Extending the surgeon's hands with tremor-free precision and minimally invasive access

1

Teleoperated systems

Surgeon-controlled manipulators (e.g. da Vinci) scale motion and filter tremor for minimally invasive surgery

2

Steerable & flexible instruments

Catheters, endoscopes, and continuum robots that navigate through natural or small-incision pathways

3

Orthopedic guidance robots

Semi-autonomous cutting/drilling within a surgeon-defined and imaging-verified boundary

4

Autonomous sub-tasks

Suturing or tissue-tracking research systems performing narrow steps under supervision



WATCH [What is the da Vinci SP robotic system?](https://www.youtube.com/watch?v=TGjnb86HndU) — Intuitive Surgical, YouTube <https://www.youtube.com/watch?v=TGjnb86HndU>

Rehabilitation & Assistive Robots

Restoring function, augmenting mobility, and supporting independent living

Robotic exoskeletons

Wearable powered limbs for gait training and daily mobility after stroke or spinal cord injury

Upper-limb rehab robots

End-effector and exoskeleton devices that guide repetitive, task-specific arm therapy

Smart prosthetics

Myoelectric or neurally controlled limbs with adaptive grip and terrain sensing

Assistive mobility & manipulation

Robotic wheelchairs and feeding/manipulation arms for people with limited motor function

ekso
Find your strength



■ **WATCH** [Find Your Strength // Ekso Bionics](https://www.youtube.com/watch?v=RtBaQEKcguk) — Ekso Bionics, YouTube <https://www.youtube.com/watch?v=RtBaQEKcguk>

Hospital Service & Logistics Robots

The unglamorous work — moved, cleaned, delivered — that frees clinical staff

Autonomous mobile robots (AMRs)

Navigate hospital corridors to move supplies, meds, linens, and lab samples

UV / disinfection robots

Autonomously sterilize rooms with UV-C light between patients

Delivery & pharmacy robots

Move medication trays and equipment between departments on demand

Telepresence units

Mobile screen-and-camera platforms for remote rounding and consults



WATCH [Meet Moxi the Robot from Diligent Robotics](https://www.youtube.com/watch?v=TASOLpNR-Gw) — Diligent Robotics, YouTube <https://www.youtube.com/watch?v=TASOLpNR-Gw>

Socially Assistive & Companion Robots

Robots whose primary function is social, emotional, or cognitive — not physical

Eldercare companions

Reminders, check-ins, and social engagement for older adults living alone

Autism & pediatric therapy robots

Structured, predictable social partners for skill-building interventions

Dementia & cognitive support

Prompting, routine support, and calming interaction for cognitive decline

Coaching & mental-health support

Conversational agents for engagement in therapy or wellness programs



■ **WATCH** [This robotic therapy seal is revolutionizing elderly care](https://www.youtube.com/watch?v=mj_-8yFxxA)
— Reuters, YouTube https://www.youtube.com/watch?v=mj_-8yFxxA

Core Technical Challenges

Every category above shares the same open problems

Safety under uncertainty

Acting near fragile, unpredictable humans with hard real-time guarantees

Human-robot interaction

Trust, transparency, and shared control between clinician/patient and machine

Perception in the clinic

Sensing tissue, people, and cluttered environments reliably and safely

Regulatory approval

Certifying a learning, physically acting system as a medical device

Data privacy & security

Protecting sensitive health data collected and acted on by autonomous systems

Cost & accessibility

Moving beyond flagship hospitals to equitable, affordable deployment

Core Technical Challenges - which is the most important for you?



Safety

0%

Human-robot interaction

0%

Perception

0%

Regulatory approval

0%

Data privacy and security

0%

Cost and accessibility

0%

Regulatory & Ethical Landscape

A healthcare robot is a medical device, a co-worker, and a data system all at once

REGULATORY

Device classification

FDA/EMA pathways (e.g. 510(k), De Novo) for autonomy levels that didn't exist when the rules were written

Liability

When a robot-assisted outcome goes wrong: manufacturer, hospital, or clinician?

Informed consent

Patients need to understand what the robot does — and what it doesn't

ETHICAL

Autonomy vs. oversight

How much decision-making should shift from clinician to system, and when?

Equity of access

Will advanced care robotics widen or narrow gaps between well- and under-resourced settings?

Dignity & deception

Especially for companion robots — does simulated empathy mislead vulnerable users?

Course Scope

01

Surgical & Interventional

Precision
manipulation inside
the body

02

Rehabilitation & Assistive

Restoring or
augmenting
movement

03

Hospital Service & Logistics

Delivery,
disinfection,
transport

04

Socially Assistive & Companion

Cognitive,
emotional, social
support

05

Diagnostic & Monitoring

Sensing, imaging,
remote exam